Q1 - 24 January - Shift 1

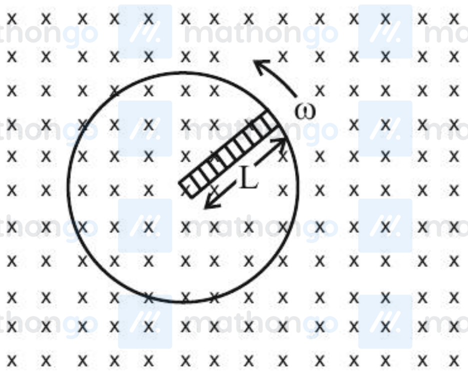
A conducting loop of radius $\frac{10}{\sqrt{\pi}}$ cm is placed perpendicular to a uniform magnetic field of 0.5T. The magnetic field is decreased to zero in 0.5 s at a steady rate. The induced emf in the circular loop at 0.25 s is:

(1) emf = 1 mV
(2) emf = 10 mV
(3) emf = 100 mV
(4) emf = 5 mV

Space for your notes:

Q2 - 24 January - Shift 2

A metallic rod of length 'L' is rotated with an angular speed of 'ω' normal to a uniform magnetic field 'B' about an axis passing through one end of rod as shown in figure. The induced emf will be :



(1) $\frac{1}{4}B^2L\omega$

(2) $\frac{1}{4}BL^2\omega$

(3) $\frac{1}{2}BL^2\omega$

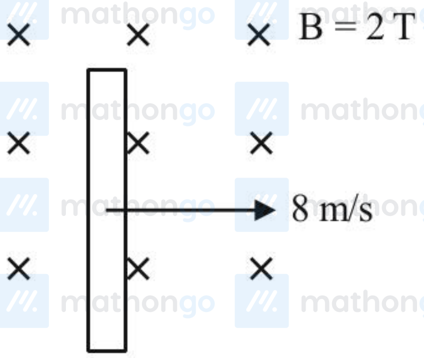
(4) $\frac{1}{2}B^2L^2\omega$

Space for your notes:

Q3 - 25 January - Shift 2

A wire of length 1 m moving with velocity 8 m/s at right angles to a magnetic field of 2T. The magnitude of induced emf, between the ends of wire will be ______ :

(1) 20 V (2) 8 V
(3) 12 V (4) 16 V



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Q4 - 29 January - Shift 1

A certain elastic conducting material is stretched into a circular loop. It is placed with its plane perpendicular to a uniform magnetic field B = 0.8 T. When released the radius of the loop starts shrinking at a constant rate of 2 cm^{-1}. The induced emf in the loop at an instant when the radius of the loop is 10 cm will be ____ mV.

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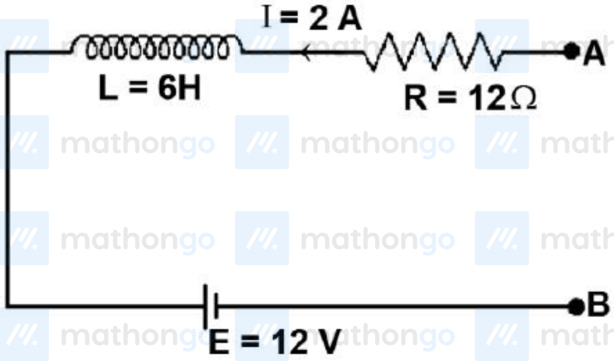
Q5 - 30 January - Shift 1

As per the given figure, if $\frac{dI}{dt} = -1$ A/s then the value of V_{AB} at this instant will be ______ V.



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Q6 - 31 January - Shift 1

An inductor of 0.5 mH, a capacitor of 20 μF and resistance of 20 Ω are connected in series with a 220 V ac source. If the current is in phase with the emf, the amplitude of current of the circuit is $\sqrt{x}$ A. The value of x is -

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2) **Q2** (3) **Q3** (4) **Q4** (10)

Q5 (30) **Q6** (242)

Q1 (2)

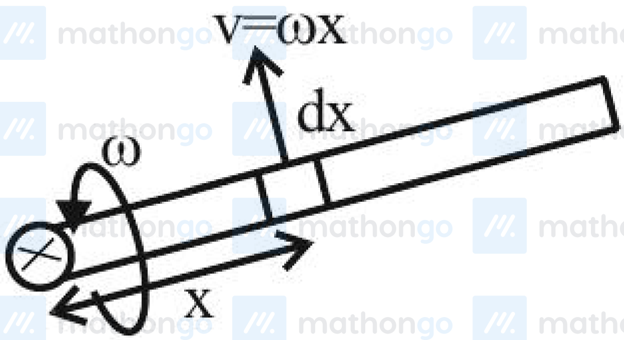
$$\text{EMF} = \frac{d\phi}{dt} = \frac{BA - 0}{t}$$

$$A = \pi r^2 = \pi\left(\frac{0.1^2}{\pi}\right) = 0.01$$

$$B = 0.5$$

$$\text{EMF} = \frac{(0.5)(0.01)}{0.5} = 0.01\text{V} = 10\text{ mV}$$

Q2 (3)



$$\int d\varepsilon = \int B(\omega x)\,dx$$

$$\varepsilon = B\omega\int_0^L x\,dx = \frac{B\omega L^2}{2}$$

Q3 (4)

Induced emf across the ends $= Bv\ell$

$= 2 \times 8 \times 1 = 16$ V

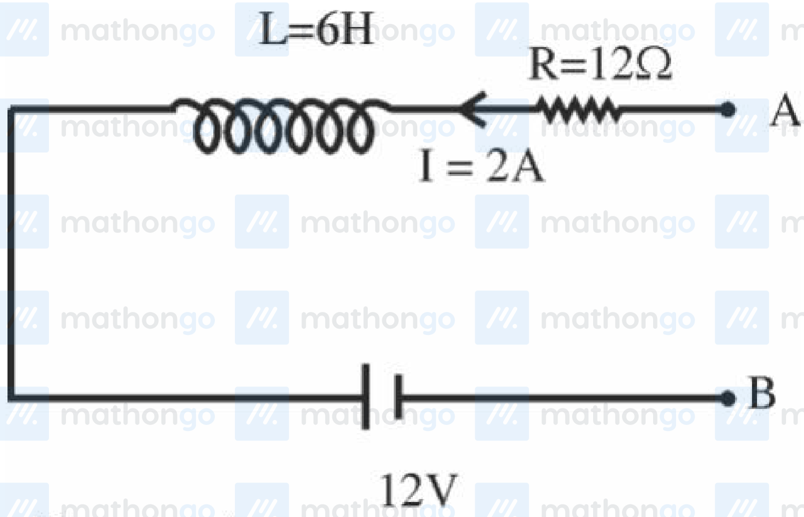
Q4 (10)

$$\text{EMF} = \frac{d}{dt}\left(B\pi r^2\right)$$

$$= 2B\pi r\frac{dr}{dt} = 2\times\pi\times 0.1\times 0.8\times 2\times 10^{-2}$$

$= 2\pi\times 1.6$ **= 10.06 [round off 10.06 = 10]**

Q5 (30)



$$\frac{dI}{dt} = -1\frac{A}{sec}$$

$$V_A - IR - L\frac{dI}{dt} - 12 = V_B$$

$$V_A - 2\times12 - 6(-1) - 12 = V_B$$

$$V_A - V_B = 36 - 6 = 30 \text{ volt}$$

Q6 (242)

$$X_L = X_C$$

$$\text{So, } Z = R = 20\ \Omega$$

$$i_{rms} = \frac{220}{20} = 11$$

$$i_{max} = 11\sqrt{2} = \sqrt{242}$$

$$\boxed{\text{Ans} : 242}$$